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Neonatal sensitivity to vocal emotions: A milestone at 37 weeks of gestational age

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Abstract

Emotional responsiveness in neonates, particularly their ability to discern vocal emotions, plays an evolutionarily adaptive role in human communication and adaptive behaviors. The developmental trajectory of emotional sensitivity in neonates is a crucial area of inquiry for understanding the foundations of early social-emotional functioning. However, the precise onset of this sensitivity in neonates and its relationship with gestational age (GA) remain subjects of investigation. In a study involving 120 healthy neonates categorized into six groups based on their GA (ranging from 35 and 40 weeks), we delved into their emotional responses to vocal stimuli. These stimuli encompassed disyllables with happy and neutral prosodies, alongside acoustically matched nonvocal control sounds. The assessments occurred during natural sleep states in neonates, utilizing the odd-ball paradigm and event-related potentials. The results unveil a distinct developmental milestone at 37 weeks GA, marking the point at which neonates exhibit heightened perceptual acuity for emotional vocal expressions. This newfound ability is substantiated by the presence of the mismatch response, akin to an initial form of adult mismatch negativity, elicited in response to positive emotional vocal prosody. Notably, this perceptual shift's specificity becomes evident when no such discrimination is observed in acoustically matched control sounds. Neonates born before 37 weeks GA do not display this level of discrimination ability. This critical developmental milestone carries significant implications for our understanding of early social-emotional development, shedding light on the role of gestational age in shaping early perceptual abilities. Moreover, it introduces the potential for a valuable screening tool in the context of autism, which is characterized by atypical social-emotional functions. This study makes a substantial contribution to the broader field of developmental neuroscience and holds promise for early intervention in neurodevelopmental disorders.

Significance statement

This study illuminates a key developmental milestone, pinpointing the emergence of heightened emotional perceptual acuity at 37 weeks of gestational age. Employing rigorous methods, we reveal that neonates at this stage exhibit remarkable discrimination abilities for emotional vocal prosody, a vital turning point in early social-emotional functioning. These findings emphasize the pivotal role of gestational age in shaping neonatal perception and provides a pathway for early screening of neurodevelopmental disorders, particularly autism. This insight holds profound implications for understanding the foundations of early social-emotional development in humans, offering a potential tool for early intervention in neurodevelopmental disorders, thereby enhancing child health and well-being.

eLife assessment

This is an **important** study on changes in newborns' neural abilities to distinguish auditory signals at 37 weeks of gestation. The evidence of change in neural discrimination as a function of gestational age is **convincing**, but further analysis of the acoustic signals and control of the infants' language environment is necessary for the results to be used in clinical applications. The work contributes to the field of neurodevelopment.

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Introduction

Emotions represent a fundamental aspect of human social interaction, serving as a compelling subject of inquiry within the disciplines of neuroscience, psychology, and psychiatry. Over the course of evolution, the human brain has evolved to possess a heightened sensitivity to the emotional expressions of others (Lindquist et al., 2012). Remarkably, even prior to the full maturation of their visual system, human infants exhibit a remarkable ability to discern vocal emotions (Blasi et al., 2011; Soderstrom et al., 2017; Vaish & Striano, 2004). Prosodic elements of speech, including pitch, intensity, and rhythm, function as universal and non-linguistic channels for emotional communication (Latinus & Belin, 2011). Numerous studies have established that infants, including those who have not yet acquired language, exhibit differentiated responses to emotional prosody conveying happiness, fear, anger, and sadness within the age range of 2 to 12 months (e.g., Caron et al., 1988; Fernald, 1993; Graham et al., 2013; Grossmann et al., 2010; Singh et al., 2002; Walker-Andrews & Grolnick, 1983; Zhao et al., 2021).

More specifically, during the very early stages of postnatal life, often termed the neonatal period (encompassing infants under four weeks of age), compelling evidence points to the presence of emotion-specific responses to emotional cues conveyed through vocal prosody. These responses have been identified through various measurement methods, including assessments of eye-opening scores (Mastropieri & Turkewitz, 1999), event-related potentials (Cheng et al., 2012), and near-infrared spectroscopy (Zhang et al., 2019). However, prior research has primarily focused on the perception and discrimination of emotions among traditionally defined term neonates, a group that includes infants born within a five-week span (37 to 41 weeks) of gestational age (GA), treating them as a homogenous cohort. This raises a crucial question: when does emotional sensitivity begin to manifest in newborns? Does it exist in preterm neonates (GA < 37 weeks)? And does it vary among neonates born at early term (GA = 37-38 weeks) and full term (GA = 39-40 weeks), as defined by the refined 'term' classification (Spong, 2013)? Surprisingly, to date, no study has explored emotion processing in neonates with varying GAs. The discovery of this developmental milestone not only advances our understanding of the cognitive mechanisms underlying human social-emotional functioning but also provides valuable insights for early diagnosis of neurodevelopmental disorders, such as autism (Jones et al., 2014; Molnar-Szakacs et al., 2021).

The principal objective of this study is to investigate emotional responses in neonates across a range of GAs, spanning from 35 to 40 weeks, and to determine whether their heightened sensitivity to emotional voices is influenced by GA. To achieve this, we utilized the odd-ball paradigm in conjunction with an event-related potential (ERP) component known as mismatch negativity (MMN) to probe the neurobiological encoding of emotional voices in the neonatal brain. MMN is an auditory ERP component that demonstrates a negative shift in response to deviant sounds when compared to standard sounds (Näätänen et al., 2007). Importantly, it can be elicited without requiring the subject's attention, making it particularly suitable for recording in young infants (Cheour et al., 1998, 2002). It is worth noting that in neonates, this ERP component often manifests as a positive response rather than the traditional MMN (e.g., Cheng et al., 2012; Chládková et al., 2021; Kostilainen et al., 2020; Richard et al., 2022; Thiede et al., 2019; Virtala et al., 2022; Winkler et al., 2003, 2009), leading many researchers to refer to it as the mismatch response (MMR) in the neonatal brain.

In our study, we exposed neonates to speech samples characterized by positive (i.e., happy) and neutral prosodies. Our selection of positive emotions over negative ones (e.g., fear, sadness, or anger) was guided not only by ethical considerations but also by previous research indicating an early preference for positive emotions in neonates (Farroni et al., 2007; Mastropieri & Turkewitz, 1999; Zhang et al., 2019; with the exception of Cheng et al., 2012). Additionally, to eliminate the possibility of neonates distinguishing emotional voices solely based on their low-level acoustic features, we included another set of control sounds. These nonvocal stimuli were meticulously matched with their vocal prosodic counterparts in terms of mean intensity and fundamental frequency (Cheng et al., 2012). Consequently, our primary objective is to pinpoint the developmental stage (i.e., the GA group) at which the discrimination between happy and neutral stimuli becomes apparent for emotional voices while remaining absent for acoustically matched control sounds.

Results

The MMR was extracted using ERP difference waves, computed by subtracting the ERP evoked by the standard stimulus (neutral sound) from the ERP evoked by the deviant stimulus (happy sound) (Näätänen et al., 2007). Brain electrical activity was recorded from the F3, F4, C3, C4, P3, and P4 sites following the international 10/20 system. However, this study primarily focused on data from electrodes F3 and F4, as the neonatal MMR exhibits a frontal distribution (Cheng et al., 2012; Cheour et al., 2002). **Figure 1** displays MMR waveforms recorded from all six electrodes.

Initially, we conducted a three-way repeated measures ANOVA on the mean MMR amplitudes (time window: 150 ms to 400 ms after sound onset) with factors including condition (vocal/nonvocal), hemisphere (left/right frontal, i.e., F3/F4) as within-subjects factors, and neonatal group (GA = 35, 36, 37, 38, 39, and 40 weeks) as the between-subjects factor. However, neither the main effect nor the interaction effects involving the hemisphere factor were statistically significant (for detailed statistics, please refer to the supplemental file subtitled "Result of the three-way ANOVA"). Consequently, we removed the hemisphere factor and averaged the MMR waveforms recorded at the F3 and F4 electrodes.

Subsequently, we performed a two-way repeated measures ANOVA with condition and group as the two factors. The main effect of stimuli was significant, $F(1,114) = 38.827$, $\eta_p^2 = 0.254$. Specifically, vocal stimuli elicited larger MMRs (mean ± standard deviation: 3.839 ± 4.855μV) compared to nonvocal stimuli (0.496 ± 4.779 μV). The main effect of group was also significant, $F(5,114) = 3.228$, $p = 0.009$, $\eta_p^2 = 0.124$. In general, MMR amplitudes were smaller in the GA35 (0.590 ± 4.579 μV) and GA36 (0.141 ± 4.807 μV) groups compared to the GA37 (2.801 ± 5.585 μV), GA38 (2.760 ± 4.382 μV), GA39 (3.401 ± 4.871 μV), and GA40 groups (3.311 ± 5.491 μV). However, no significant differences were found in pairwise comparisons after Bonferroni adjustment for multiple comparisons.

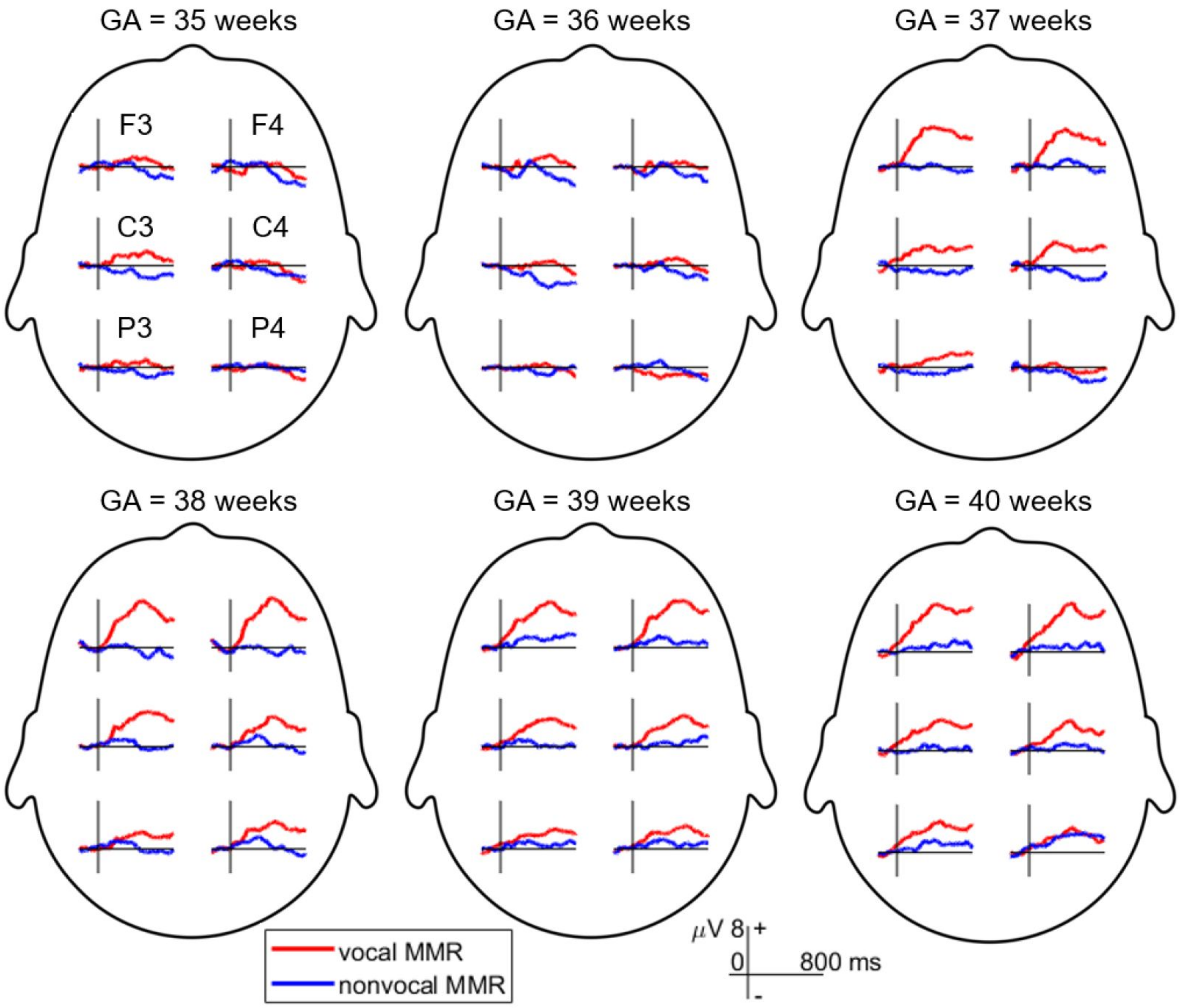

Figure 1.

Mismatch response (MMR) waveforms recorded across six electrodes in different gestational age (GA) groups. The MMR is extracted by subtracting the event-related potential (ERP) elicited by the standard stimulus (neutral condition) from the ERP evoked by the deviant stimulus (happy condition).

The interaction between stimuli and group was significant, $F(5,114) = 3.127$, $p = 0.011$, $\eta_p^2 = 0.121$ (as shown in **Figure 2**). Simple effect analysis revealed that MMR amplitudes were larger in the vocal condition compared to the nonvocal condition in the GA37 ($F(1,114) = 15.254$, $p < 0.001$, $\eta_p^2 = 0.118$; vocal = 5.367 ± 5.165 μV, nonvocal = 0.235 ± 4.847 μV), GA38 ($F(1,114) = 16.072$, $p < 0.001$, $\eta_p^2 = 0.124$; vocal = 5.394 ± 3.145 μV, nonvocal = 0.126 ± 3.861 μV), GA39 ($F(1,114) = 8.393$, $p = 0.005$, $\eta_p^2 = 0.069$; vocal = 5.305 ± 4.011 μV, nonvocal = 1.498 ± 4.998 μV), and GA40 groups ($F(1,114) = 14.482$, $p < 0.001$, $\eta_p^2 = 0.113$; vocal = 5.811 ± 5.298 μV, nonvocal = 0.811 ± 4.546 μV). However, there were no significant differences in MMR amplitudes between the two kinds of stimuli in the GA35 ($F(1,114) = 0.026$, $\eta_p^2 < 0.001$; ; vocal = 0.695 ± 4.031 μV, nonvocal = 0.485 ± 5.173 μV) and GA36 groups ($F(1,114) = 0.236$, $p = 0.628$, $\eta_p^2 = 0.002$; vocal = 0.460 ± 4.104 μV, nonvocal = -0.179 ± 5.511 μV).

Further analysis revealed that vocal stimuli evoked varying MMR amplitudes across groups, $F(5,114) = 6.768$, $p < 0.001$, $\eta_p^2 = 0.229$. Specifically, the MMRs evoked by vocal stimuli were smaller in the GA35 group compared to GA37 ($p = 0.014$), GA38 ($p = 0.013$), GA39 ($p = 0.017$), and GA40 groups ($p = 0.005$). Similarly, the MMRs evoked by vocal stimuli were smaller in the GA36 group compared to GA37 ($p = 0.008$), GA38 ($p = 0.008$), GA39 ($p = 0.009$), and GA40 groups ($p = 0.003$). However, nonvocal stimuli did not elicit significantly different MMR amplitudes across groups, $F(5,114) = 0.300$, $p = 0.912$, $\eta_p^2 = 0.013$.

Discussion

The current study elucidates a pivotal developmental milestone in neonatal emotional responsiveness by investigating their ability to perceive vocal emotions. The findings illuminate a distinct turning point at 37 weeks of gestational age (GA), representing the onset of heightened perceptual acuity for emotional vocal expressions. This milestone is particularly evident in the robust MMR to positive emotional vocal prosody. Significantly, the absence of this discrimination ability when acoustically matched control sounds were presented underscores the specificity of this developmental shift towards emotional voice processing. Our identification of the 37-week GA mark aligns with previous research, which indicated emotional sensitivity in term neonates born at or after 37 weeks of gestation (Cheng et al., 2012; Farroni et al., 2007; Mastropieri & Turkewitz, 1999; Zhang et al., 2019) and in preterm neonates (GA < 37 weeks) tested at term age (Kostilainen et al., 2020). Notably, our findings also reveal that neonates born before 37 weeks GA do not exhibit these emotional discrimination abilities.

In the final trimester of pregnancy, the human brain undergoes a period of rapid and continuous changes in neural structure and cognitive functions (Bayer et al., 1993; Clancy et al., 2007). Although there is no direct evidence to support the notion that preterm neonates can decode vocal emotions, they have displayed an aptitude for processing speech stimuli with social contexts. For example, neonates born at or after 29 weeks GA have shown a preference for infant-directed speech, characterized by a high pitch, exaggerated pitch modifications, and a slow rate (Butler et al., 2014). This preference has been associated with increased visual attention, heightened alertness (Eckerman et al., 1994), reduced heart rate (White-Traut et al., 1997), and enhanced speech differentiation in premature babies (Richard et al., 2022). Additionally, neonates born at or after 30 weeks GA have demonstrated an age-related increase in sensitivity to maternal voices (D Chorna et al., 2018; A. P. F. Key et al., 2012), leading to beneficial effects on cognitive and neurobehavioral development (Caskey et al., 2011; Picciolini et al., 2014; see Provenzi et al., 2018 for a review). These effects encompass improved feeding behaviors, heightened responsiveness (Katz, 1971; Krueger et al., 2010), enhanced weight gain (Zimmerman et al., 2013), and activated auditory cortical plasticity (Webb et al., 2015). Both infant-directed speech and maternal voices feature extensive pitch modulation, and the preference for these emotionally prosodic-like voices during the preterm stage may prepare the developing brain to discriminate vocal emotions at 37 weeks GA, as demonstrated in this study.

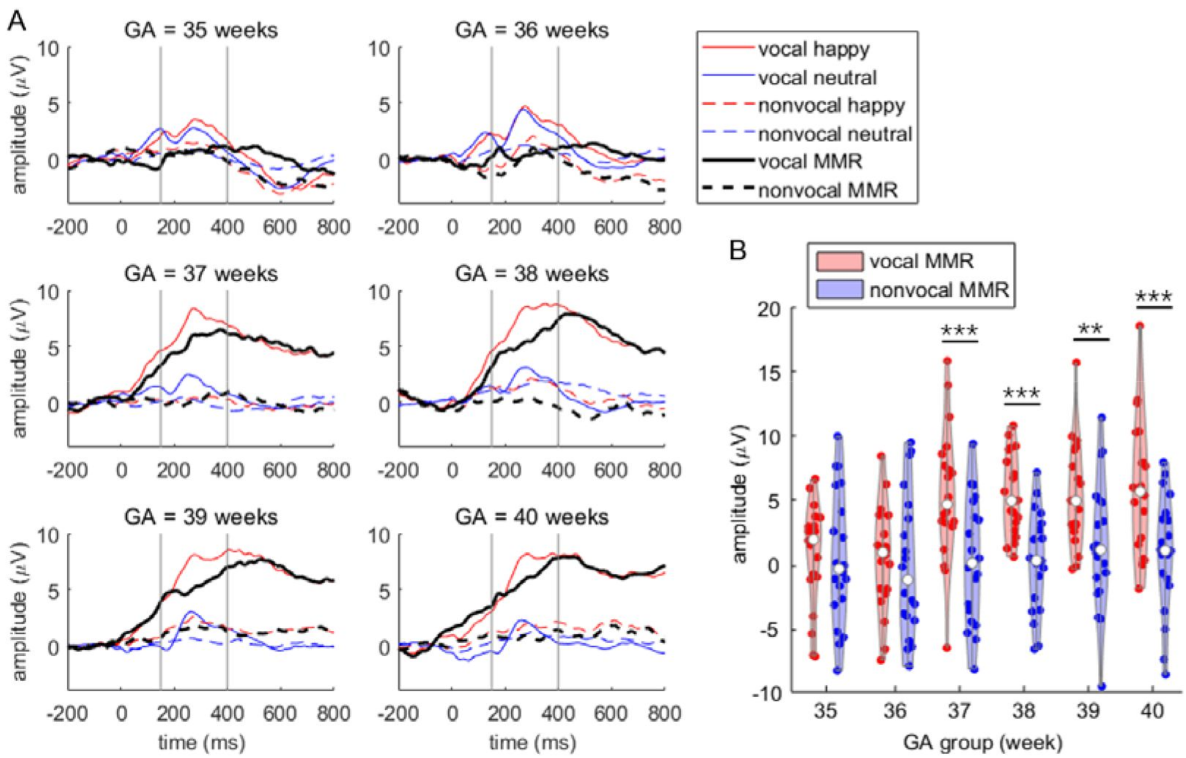

Figure 2.

Primary MMR results. A, ERP waveforms (averaged at the F3 and F4 electrodes) across the six GA groups. The time window for assessing MMR amplitude is indicated between the two vertical gray lines. B, Violin plot illustrating MMR amplitudes in the six GA groups. Simple effect analysis was performed with pairwise comparisons, corrected using the Bonferroni method: ** $p < 0.010$, *** $p < 0.001$.

Traditionally, 37 weeks of gestation served as the benchmark for fetal maturity, and term infants born within the 37 to 41 weeks GA range were generally considered healthy, forming a homogenous group. Recent insights, however, have unveiled variations in physical and cognitive maturation within this 5-week span of full-term pregnancy. Research indicates that neonates born at 37-38 weeks GA face increased risks of neonatal mortality and pediatric respiratory, neurologic, and endocrine morbidities compared to those born at 39-41 weeks GA (Cahen-Peretz et al., 2022; Clark et al., 2009; Edwards et al., 2013; Ghartey et al., 2012; Paz Levy et al., 2017; Sengupta et al., 2013; Tita et al., 2009). Furthermore, a dose-response relationship inversely linking GA to the risk of developmental delay has been identified in infants from preterm to full-term births (Rose et al., 2013; Schonhaut et al., 2015). Early birth (34-38 weeks GA) has also been found to have a detrimental impact on child development and academic achievement during school age (Bentley et al., 2016; Chan et al., 2016; Dong et al., 2012; Hedges et al., 2021; Murray et al., 2017; Nielsen et al., 2019; Noble et al., 2012). Consequently, the definition of a full-term pregnancy has been narrowed to a two-week window starting at 39 weeks (Spong, 2013), with nonmedically indicated deliveries between 37 and 38 weeks of gestation discouraged (ACOG Committee Opinion, 2019). While accumulating evidence underscores the adverse effects of the traditional 37-week threshold, our findings contribute to a limited body of research suggesting that neonatal social-emotional functioning reaches a development milestone at 37 weeks GA.

A more comprehensive understanding of the developmental trajectory of emotional sensitivity has the potential to revolutionize decision-making in the final weeks of pregnancy and the identification of newborns at risk of emotional and neurodevelopmental disorders, particularly autism. Individuals with autism often exhibit atypical perceptual and neural processing of emotional information, including emotional prosodic voices (Kuhl et al., 2005; Lindström et al., 2018; Van Lancker et al., 1989; Wang et al., 2007; for comprehensive reviews, see Frühholz & Staib, 2017; Yeung, 2022). While previous studies have indicated that social-emotional behavioral indicators typically begin to demonstrate predictive power for autism from the second year of life (Gliga et al., 2014; Jones et al., 2014), brain functional indicators of emotional processing during infancy, especially within the first year of life, have already shown their predictive value (Ayoub et al., 2022; Clairmont et al., 2021; Molnar-Szakacs et al., 2021). For instance, infants subsequently diagnosed with autism displayed a smaller amplitude and shorter duration of the negative central (Nc) component at six months of age when viewing smiling faces compared to toys, a pattern not observed in infants who were subsequently undiagnosed (Jones et al., 2016). Additionally, while the Nc and P400 components were able to distinguish between smiling, fearful, and neutral facial expressions in typically developing 9-to-10-month-old infants, these EEG indicators failed to differentiate emotional faces in infants at high risk for autism (Di Lorenzo et al., 2021; A. P. Key et al., 2015). Moreover, it has been observed that infants at high risk for autism exhibit diminished activation in the fusiform gyrus and hippocampus compared to healthy controls when exposed to sad cries between the ages of 4 and 7 months (Blasi et al., 2015). The fusiform gyrus, a region crucial for face perception and memory, and the hippocampus, which plays a significant role in general learning and memory processes, are both implicated in this phenomenon (Lisman et al., 2017; Rossion et al., 2024). Consequently, the hippocampus-fusiform network, essential for the development of social cognitive skills, may serve as a predictive indicator for the onset of autism. Building upon these existing studies, our research introduces a promising early screening indicator for autism: the neonatal MMR in response to emotional voices. We recommend future longitudinal studies to further elucidate the predictive role of this neurophysiological indicator, thereby facilitating early diagnosis and intervention for social-emotional disorders.

When interpretating the current findings, it is important to consider that the nonvocal control sounds utilized in this study may not have adequately eliminated all low-level acoustic properties that could aid neonatal discrimination. Specifically, while the nonvocal control counterparts retained the fundamental frequency (f0) of the emotional prosodic voices, they did not replicate the burst of energy associated with consonants. Consequently, it cannot be ruled out that neonates utilized consonant characteristics to discriminate emotional prosodies conveyed by disyllables. Additionally, the nonvocal sounds were generated using a simple filtering method, resulting in certain vocal-like components persisting in these control sounds. The limitations of the control sound materials should be given greater consideration in future replication or further research.

Furthermore, there is a compelling need for future investigations to expand upon the present findings by incorporating a broader array of emotional stimuli. Non-speech emotional vocalizations, such as laughter, crying, or retching, as well as natural emotional auditory cues like thunder, flowing water, hissing snakes, and bird calls, offer a rich spectrum of emotional materials that have been shown to engage the perceptual faculties of neonates and infants (Blasi et al., 2011; Erlich et al., 2013). This multifaceted approach could illuminate whether the developmental milestone observed at 37 weeks GA is specific to the processing of emotional prosodic speech and vocal expressions, or if it extends to encompass a broader range of both artificial and natural emotional auditory cues. Moreover, the use of non-speech emotional stimuli aids in resolving the debate between nature and nurture concerning the onset of emotional sensitivity at 37 weeks GA. It cannot solely attribute the current finding of discrimination to the innate maturational explanation, given that the auditory system becomes functional at the end of the second trimester of pregnancy, allowing exposure to spoken language in utero to influence the development of speech perception (DeCasper & Spence, 1986; Moon et al., 2013; Partanen et al., 2013). The ability to discriminate prosodic emotions starting at 37 weeks GA could stem from additional exposure in utero to speech. Future exploration is need to definitively investigate prenatal learning by utilizing emotional sounds that are infrequently encountered in the prenatal environment. Finally, the inclusion of non-speech emotional materials may offer insights into the potential right lateralization of emotional processing in the neonatal brain. While prior studies (including some cited herein) have identified right lateralization for emotional processing in full-term neonates (Cheng et al., 2012; Zhang et al., 2019; see Bisiacchi & Cainelli, 2022 for a comprehensive review), the introduction of non-speech materials can help disentangle the confounding effects of left lateralization, which is associated with language processing and has been identified in both preterm (Mahmoudzadeh et al., 2013) and full-term neonates (Kotilahti et al., 2010; May et al., 2018; Peña et al., 2003; Sato et al., 2012; Vannasing et al., 2016; Wu et al., 2022).

In summary, this study has illuminated a pivotal developmental milestone – the emergence of heightened perceptual acuity for emotional vocal expressions at 37 weeks GA. It is important to note that neonates' perceptual sensitivity at this stage is unlikely to be associated with a profound conceptual understanding of the meaning of emotions. Nevertheless, this unique discrimination ability in early life may serve as a foundational building block for the subsequent development of emotional and social cognition. Beyond its scientific significance, our findings underscore the critical role of gestational age in shaping early perceptual abilities and offer a promising avenue for early screening and intervention in neurodevelopmental disorders, particularly autism, where early detection is of paramount importance. This work not only deepens our understanding of neonatal social and emotional development but also provides a potential tool to support early diagnosis and intervention in this critical realm of child health and well-being.

Materials and methods

Subjects

The research received approval from both the Ethical Committee of Peking University First Hospital and the Chinese Clinical Trial Registry (ChiCTR2300069898). Initially, we planned to include 120 healthy neonates, with 60 being boys, in the data analysis. These participants were categorized into six groups based on their GA, specifically 35, 36, 37, 38, 39, and 40 weeks, with each group comprising twenty subjects. For instance, the GA35 group comprised neonates with GA ranging from 35 weeks plus 0 day to 6 days. However, we ultimately recruited 198 neonates to obtain 120 valid datasets due to non-cooperation of newborns (n = 75) or technical issues (n = 3). Specially, 11, 12, 11, 14, 13, and 14 neonates were excluded from data analysis in the GA35, GA36, GA37, GA38, GA39, and GA40 groups, respectively, due to crying or irritable movements during EEG device preparation and EEG recording.

The mothers of these neonates were monolingual and nurtured their babies in a native language environment. All neonates participated the experiment within the first 24 hours after birth, with a mean ± standard deviation of 17.8 ± 0.4 hours for the 120 valid data.

Prior to data collection, written consent was obtained from the parents or legal guardians of all participating neonates for access to clinical information and EEG data collection for scientific purposes. While sample sizes were not statistically predetermined, including twenty subjects per GA group represented the maximum feasible number within a two-year period at Peking University First Hospital.

All subjects met the following inclusion criteria: 1) normal birth weight for their GA; 2) absence of clinical symptoms at the time of EEG recording; 3) no previous sedation or medication prior to EEG recording; and 4) normal hearing results in an evoked otoacoustic emissions test (ILO88 Dpi, Otodynamics Ltd, Hatfield, UK). Additionally, subjects did not exhibit any of the following neurological or metabolic disorders: 1) hypoxic-ischemic encephalopathy, 2) intraventricular hemorrhage or white matter damage detected by cranial ultrasound, 3) congenital malformation, 4) central nervous system infection, 5) metabolic disorder, 6) clinical evidence of seizures, and 7) signs of asphyxia.

Stimuli

A total of 85 possible combinations of consonants and vowels, which are standard in Chinese (Lee & Zee, 2003) and common to most human languages (e.g., 'dada' and 'keke'), were recorded by a native Chinese-speaking adult woman with the Peking dialect. Each disyllable was recorded with four repetitions, two using a happy prosody and two with a neutral prosody, resulting in a total of 340 disyllables (85 × 4). Twenty Chinese undergraduate students (10 men, mean age 20.1 ± 1.2 years) performed a discrimination task, distinguishing between happy and neutral stimuli, and rated the affective content of these stimuli.

In the affective rating task, participants assessed the intensity of happiness (on a 9-point scale ranging from 1 being the least happy to 9 being the happiest) and the valence (on a 9-point scale ranging from 1 being the most negative, 5 being neutral, to 9 being the most positive) of the 340 stimuli. This study selected five pairs of happy and neutral disyllables that shared the same consonant-monophthong combinations and achieved 100% discrimination accuracy in the discrimination task (i.e., 'dada', 'dudu', 'gege', 'keke', and 'tutu' in Chinese Pinyin). Paired-samples t-tests demonstrated that the happy disyllables were rated as significantly happier ($t(4) = 24.70$, $p < 0.001$; happy intensity: 7.49 ± 0.20 versus 3.53 ± 0.19) and had a more positive valence ($t(4) = 18.55$, $p < 0.001$; valence: 7.11 ± 0.12 versus 4.91 ± 0.24) than their neutral counterparts. These ten disyllables were then standardized to have the same mean intensity and had duration of 400 ms using Adobe Audition (v.2022; Adobe Systems Inc., San Jose, CA).

To ensure that neonates were discriminating based on prosodic cues containing emotional content rather than low-level acoustic properties, we employed a method similar to Cheng et al. (2012) and created a separate set of nonvocal control sounds. We hypothesized that the fundamental frequency (f0) alone does not convey emotional content in voices and that neonates require multiple other prosodic cues embedded in the high-frequent component to discern emotions, as suggested by Cheng et al. (2012) and Zhang et al. (2014). As a result, ten nonvocal sounds were generated to match the f0 contours and temporal envelopes of their corresponding vocal sounds. This matching process was carried out using Matlab (v.2021b; MathWorks, Inc., South

Natick, MA). Specifically, we initially applied a zero-phase filter with a bandpass of mean f0 ± 150 Hz to obtain f0-matched sounds of prosodic voices. Subsequently, a normalization procedure was implemented to ensure that the intensity of each pair of vocal and nonvocal sounds was equal. Oscillograms and spectrograms of the auditory stimuli utilized in this study are presented in **Figure 3**, generated using Praat (v.6.3.17, *www.praat.org*). All auditory stimuli, along with their pronunciations and rating scores, are available in the supplemental material labeled "experimental sounds".

To optimize the diversity of our material and increase the generalizability of our results, we utilized ten sets of sounds. Each set included both positive and neutral prosodic voices, along with their respective nonvocal counterparts. These auditory materials were distributed randomly and evenly within each neonatal GA group, ensuring that each set was presented twice (to two individuals) in each GA group.

Procedure

The sound stimuli were presented in two blocks: the vocal and nonvocal conditions, utilizing the odd-ball paradigm. The standard stimulus was either a vocal or nonvocal neutral sound, while the deviant stimulus was either a vocal or nonvocal happy sound. Each block consisted of 240 standard stimuli (80%) and 60 deviant stimuli (20%). The standard and deviant stimuli were presented randomly, ensuring that each deviant stimulus was followed by at least two standard stimuli. Each sound had a duration of 400 ms, and the inter-trial interval was silent, with varying durations ranging from 500 to 700 ms. Each block lasted for 5 minutes, and the order of the vocal and nonvocal blocks was counterbalanced across participants. A 5-minute break separated the two blocks, resulting in a total EEG recording duration of 15 minutes.

The experiment took place in the neonatal ward of Peking University First Hospital. Neonates were transported to a designated testing room for EEG recording as soon as their condition stabilized after birth. In this room, they were separated from their mothers to minimize any natural exposure to speech or speech stimuli other than those utilized in the experiment. Auditory stimuli were presented through a pair of loudspeakers positioned approximately 30 cm away from the neonates' left and right ears, at a sound pressure level of 55 to 60 dB, with an average background noise intensity level of 30 dB. EEG recording was conducted while the neonates were in a natural sleep state (Cheour et al., 2002; Wu et al., 2022).

Data recording and analysis

We recorded brain electrical activity using an electrical amplifier (NeuSen.W32, Neuracle, Changzhou, China) at a sampling frequency of 1000 Hz. Initially, the data were recorded online with reference to the left mastoid and subsequently re-referenced offline to the average of the left and right mastoids. The ground electrode was positioned on the forehead. For the recording of vertical eye movements, an electrooculogram (EOG) electrode was positioned beneath the left eye, while another was placed at the left external canthi for recording horizontal eye movements. Throughout the recording process, electrode impedances were meticulously maintained below 10 kΩ.

We eliminated ocular artifacts from the EEG data using a regression procedure implemented in NeuroScan software (Scan 4.3, NeuroScan, Herndon, VA). Subsequently, we employed Matlab (v.2021b; Mathworks, Inc., Sherborn, MA) for data processing and result presentation. The EOG-corrected EEG data were then offline filtered with a half-amplitude cutoff range of 0.01~30 Hz and segmented from 200 ms before sound presentation until 1000 ms after sound onset. Epochs were baseline-corrected relative to the mean voltage during the 200 ms preceding sound presentation. Any epochs containing artifacts with peak deflections exceeding ±200 μV were rejected (see also Biro et al., 2021; Di Lorenzo et al., 2021; Kumaravel et al., 2022), followed by averaging for each experimental condition. The number of valid epochs did not exhibit a significant difference

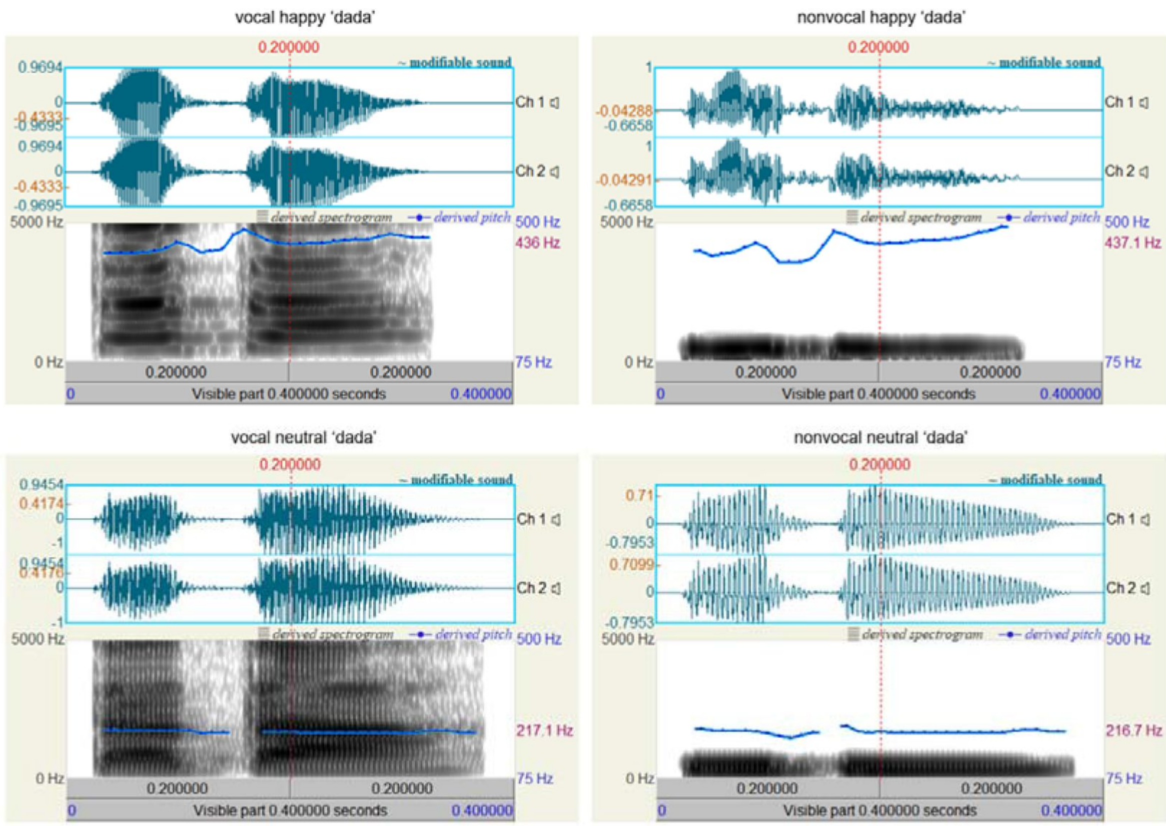

Figure 3.

Oscillograms and spectrograms displaying vocal and nonvocal auditory stimuli for the syllable 'dada' (international phonetic symbol: [ta ta]).

across neonatal groups (please refer to the supplemental file subtitled “Epoch number”). The time window for the MMR component was pre-defined as 150 ms to 400 ms after sound onset, based on prior knowledge (Cheour et al., 2002), and utilized throughout the data analysis.

We performed statistical analyses using SPSS Statistics (v. 20.0; IBM, Somers, USA). Descriptive data are reported as mean ± standard deviation. The significance level was set at 0.05. We applied the Greenhouse-Geisser correction for ANOVA tests when deemed appropriate. Post-hoc tests for significant main effects were conducted using the Bonferroni method. Significant interactions were explored through simple effects models. We reported partial eta-squared (η_p^2) as a measure of effect size in ANOVA tests.

Acknowledgements

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Data availability

The experimental materials are accessible as supplementary files for download. EEG epochs from all 120 datasets can be downloaded from *https://osf.io/a3xzy/*. The data and materials from this study are available for academic purpose free of charge, on the condition that proper citation to this article is provided.

Author contributions

D. Zhang and X. Hou conceptualized and designed the research. P. Zhang, X. Hou, and C. Peng conducted the experiments, while P. Zhang, L. Mo, and D. Zhang were responsible for data analysis. D. Zhang authored the paper, and X. Hou revised and provided critical input to the manuscript.

References

American College of Obstetricians and Gynecologists, & Society for Maternal-Fetal Medicine (2019) **ACOG committee opinion no .765: Avoidance of nonmedically indicated early-term deliveries and associated neonatal morbidities** *Obstetrics & Gynecology* **133**:e156–e163 https://doi.org/10.1097/AOG.0000000000003076

Ayoub M. J., Keegan L., Tager-Flusberg H., Gill S. V. (2022) **Neuroimaging techniques as descriptive and diagnostic tools for infants at risk for autism spectrum disorder: A systematic review** *Brain Sciences* **12** https://doi.org/10.3390/brainsci12050602

Bayer S. A., Altman J., Russo R. J., Zhang X. (1993) **Timetables of neurogenesis in the human brain based on experimentally determined patterns in the rat** *Neurotoxicology* **14**:83–144

Bentley J. P., Roberts C. L., Bowen J. R., Martin A. J., Morris J. M., Nassar N. (2016) **Planned birth before 39 weeks and child development: A population-based study** *Pediatrics* **138** https://doi.org/10.1542/peds.2016-2002

Biro S., Peltola M. J., Huffmeijer R., Alink L. R. A., Bakermans-Kranenburg M. J., van IJzendoorn M. H. (2021) **Frontal EEG asymmetry in infants observing separation and comforting events: The role of infants' attachment relationship** *Developmental Cognitive Neuroscience* **48** https://doi.org/10.1016/j.dcn.2021.100941

Bisiacchi P., Cainelli E. (2022) **Structural and functional brain asymmetries in the early phases of life: A scoping review** *Brain Structure and Function* **227**:479–496 https://doi.org/10.1007/s00429-021-02256-1

Blasi A., Lloyd-Fox S., Sethna V., Brammer M. J., Mercure E., Murray L., Williams S. C. R., Simmons A., Murphy D. G. M., Johnson M. H. (2015) **Atypical processing of voice sounds in infants at risk for autism spectrum disorder** *Cortex* **71**:122–133 https://doi.org/10.1016/j.cortex.2015.06.015

Blasi A. *et al.* (2011) **Early specialization for voice and emotion processing in the infant brain** *Current Biology* **21**:1220–1224 https://doi.org/10.1016/j.cub.2011.06.009

Butler S. C., O'Sullivan L. P., Shah B. L., Berthier N. E. (2014) **Preference for infant-directed speech in preterm infants** *Infant Behavior and Development* **37**:505–511 https://doi.org/10.1016/j.infbeh.2014.06.007

Cahen-Peretz A., Tsaitlin-Mor L., Abu-Ahmad W., Ben-Shushan M. T., Levine H., Walfisch A. (2022) **Long-term respiratory outcomes in early-term born offspring: A systematic review and meta-analysis** *American Journal of Obstetrics and Gynecology MFM* **4** https://doi.org/10.1016/j.ajogmf.2022.100570

Caron A. J., Caron R. F., MacLean D. J. (1988) **Infant discrimination of naturalistic emotional expressions: The role of face and voice** *Child Development* **59**:604–616

Caskey M., Stephens B., Tucker R., Vohr B. (2011) **Importance of parent talk on the development of preterm infant vocalizations** *Pediatrics* **128**:910–916 https://doi.org/10.1542/peds.2011-0609

Chan E., Leong P., Malouf R., Quigley M. A. (2016) **Long-term cognitive and school outcomes of late-preterm and early-term births: A systematic review** *Child: Care, Health and Development* **42**:297–312 https://doi.org/10.1111/cch.12320

Cheng Y., Lee S.-Y., Chen H.-Y., Wang P.-Y., Decety J. (2012) **Voice and emotion processing in the human neonatal brain** *Journal of Cognitive Neuroscience* **24**:1411–1419 https://doi.org/10.1162/jocn_a_00214

Cheour M., Ceponiene R., Lehtokoski A., Luuk A., Allik J., Alho K., Näätänen R. (1998) **Development of language-specific phoneme representations in the infant brain** *Nature Neuroscience* **1**:351–353 https://doi.org/10.1038/1561

Cheour M. *et al.* (2002) **Speech sounds learned by sleeping newborns** *Nature* **415**:599–600 https://doi.org/10.1038/415599b

Chládková K., Urbanec J., Skálová S., Kremláček J. (2021) **Newborns' neural processing of native vowels reveals directional asymmetries** *Developmental Cognitive Neuroscience* **52** https://doi.org/10.1016/j.dcn.2021.101023

Clairmont C., Wang J., Tariq S., Sherman H. T., Zhao M., Kong X.-J. (2021) **The value of brain imaging and electrophysiological testing for early screening of autism spectrum disorder: A systematic review** *Frontiers in Neuroscience* **15** https://doi.org/10.3389/fnins.2021.812946

Clancy B., Finlay B. L., Darlington R. B., Anand K. J. S. (2007) **Extrapolating brain development from experimental species to humans** *Neurotoxicology* **28**:931–937 https://doi.org/10.1016/j.neuro.2007.01.014

Clark S. L., Miller D. D., Belfort M. A., Dildy G. A., Frye D. K., Meyers J. A. (2009) **Neonatal and maternal outcomes associated with elective term delivery** *American Journal of Obstetrics and Gynecology* **200** https://doi.org/10.1016/j.ajog.2008.08.068

D Chorna O., L Hamm E., Shrivastava H., Maitre N. L. (2018) **Feasibility of event-related potential (ERP) biomarker use to study effects of mother's voice exposure on speech sound differentiation of preterm infants** *Developmental Neuropsychology* **43**:123–134 https://doi.org/10.1080/87565641.2018.1433671

DeCasper A. J., Spence M. J. (1986) **Prenatal maternal speech influences newborns' perception of speech sounds** *Infant Behavior and Development* **9**:133–150 https://doi.org/10.1016/0163-6383(86)90025-1

Di Lorenzo R., Munsters N. M., Ward E. K., de Jonge M., Kemner C., van den Boomen C. (2021) **Is it fear? Similar brain responses to fearful and neutral faces in infants with a heightened likelihood for autism spectrum disorder** *Journal of Autism and Developmental Disorders* **51**:961–972 https://doi.org/10.1007/s10803-020-04560-x

Dong Y., Chen S., Yu J. (2012) **A systematic review and meta-analysis of long-term development of early term infants** *Neonatology* **102**:212–221 https://doi.org/10.1159/000338099

Eckerman C. O., Oehler J. M., Medvin M. B., Hannan T. E. (1994) **Premature newborns as social partners before term age** *Infant Behavior and Development* **17**:55–70 https://doi.org/10.1016/0163-6383(94)90022-1

Edwards M. O., Kotecha S. J., Kotecha S. (2013) **Respiratory distress of the term newborn infant** *Paediatric Respiratory Reviews* **14**:29–36 https://doi.org/10.1016/j.prrv.2012.02.002

Farroni T., Menon E., Rigato S., Johnson M. H. (2007) **The perception of facial expressions in newborns** *The European Journal of Developmental Psychology* **4**:2–13 https://doi.org/10.1080/17405620601046832

Fernald A. (1993) **Approval and disapproval: Infant responsiveness to vocal affect in familiar and unfamiliar languages** *Child Development* **64**:657–674

Frühholz S., Staib M. (2017) **Neurocircuitry of impaired affective sound processing: A clinical disorders perspective** *Neuroscience and Biobehavioral Reviews* **83**:516–524 https://doi.org/10.1016/j.neubiorev.2017.09.009

Ghartey K., Coletta J., Lizarraga L., Murphy E., Ananth C. V., Gyamfi-Bannerman C. (2012) **Neonatal respiratory morbidity in the early term delivery** *American Journal of Obstetrics and Gynecology* **207** https://doi.org/10.1016/j.ajog.2012.07.022

Gliga T., Jones E. J. H., Bedford R., Charman T., Johnson M. H. (2014) **From early markers to neuro-developmental mechanisms of autism** *Developmental Review* **34**:189–207 https://doi.org/10.1016/j.dr.2014.05.003

Graham A. M., Fisher P. A., Pfeifer J. H. (2013) **What sleeping babies hear: A functional MRI study of interparental conflict and infants' emotion processing** *Psychological Science* **24**:782–789 https://doi.org/10.1177/0956797612458803

Grossmann T., Oberecker R., Koch S. P., Friederici A. D. (2010) **The developmental origins of voice processing in the human brain** *Neuron* **65**:852–858 https://doi.org/10.1016/j.neuron.2010.03.001

Hedges A., Corman H., Noonan K., Reichman N. E. (2021) **Gestational age at term and educational outcomes at age nine** *Pediatrics* **148** https://doi.org/10.1542/peds.2020-021287

Jones E. J. H., Gliga T., Bedford R., Charman T., Johnson M. H. (2014) **Developmental pathways to autism: A review of prospective studies of infants at risk** *Neuroscience and Biobehavioral Reviews* **39**:1–33 https://doi.org/10.1016/j.neubiorev.2013.12.001

Jones E. J. H., Venema K., Earl R., Lowy R., Barnes K., Estes A., Dawson G., Webb S. J. (2016) **Reduced engagement with social stimuli in 6-month-old infants with later autism spectrum disorder: A longitudinal prospective study of infants at high familial risk** *Journal of Neurodevelopmental Disorders* **8** https://doi.org/10.1186/s11689-016-9139-8

Katz V. (1971) **Auditory stimulation and developmental behavior of the premature infant** *Nursing Research* **20**

Key A. P. F., Lambert E. W., Aschner J. L., Maitre N. L. (2012) **Influence of gestational age and postnatal age on speech sound processing in NICU infants** *Psychophysiology* **49**:720–731 https://doi.org/10.1111/j.1469-8986.2011.01353.x

Key A. P., Ibanez L. V., Henderson H. A., Warren Z., Messinger D. S., Stone W. L. (2015) **Positive affect processing and joint attention in infants at high risk for autism: An exploratory study** *Journal of Autism and Developmental Disorders* **45**:4051–4062 https://doi.org/10.1007/s10803-014-2191-x

Kostilainen K., Partanen E., Mikkola K., Wikström V., Pakarinen S., Fellman V., Huotilainen M. (2020) **Neural processing of changes in phonetic and emotional speech sounds and tones in preterm infants at term age** *International Journal of Psychophysiology* **148**:111–118 https://doi.org/10.1016/j.ijpsycho.2019.10.009

Kotilahti K., Nissilä I., Näsi T., Lipiäinen L., Noponen T., Meriläinen P., Huotilainen M., Fellman V. (2010) **Hemodynamic responses to speech and music in newborn infants** *Human Brain Mapping* **31**:595–603 https://doi.org/10.1002/hbm.20890

Krueger C., Parker L., Chiu S.-H., Theriaque D. (2010) **Maternal voice and short-term outcomes in preterm infants** *Developmental Psychobiology* **52**:205–212 https://doi.org/10.1002/dev.20426

Kuhl P. K., Coffey-Corina S., Padden D., Dawson G. (2005) **Links between social and linguistic processing of speech in preschool children with autism: Behavioral and electrophysiological measures** *Developmental Science* **8**:F1–F12 https://doi.org/10.1111/j.1467-7687.2004.00384.x

Kumaravel V. P., Farella E., Parise E., Buiatti M. (2022) **NEAR: An artifact removal pipeline for human newborn EEG data** *Developmental Cognitive Neuroscience* **54** https://doi.org/10.1016/j.dcn.2022.101068

Latinus M., Belin P. (2011) **Human voice perception** *Current Biology* **21**:R143–145 https://doi.org/10.1016/j.cub.2010.12.033

Lee W.-S., Zee E. (2003) **Standard Chinese (Beijing)** *Journal of the International Phonetic Association* **33**:109–112 https://doi.org/10.1017/S0025100303001208

Lindquist K. A., Wager T. D., Kober H., Bliss-Moreau E., Barrett L. F. (2012) **The brain basis of emotion: A meta-analytic review** *The Behavioral and Brain Sciences* **35**:121–143 https://doi.org/10.1017/S0140525X11000446

Lindström R., Lepistö-Paisley T., Makkonen T., Reinvall O., Nieminen-von Wendt T., Alén R., Kujala T. (2018) **Atypical perceptual and neural processing of emotional prosodic changes in children with autism spectrum disorders** *Clinical Neurophysiology* **129**:2411–2420 https://doi.org/10.1016/j.clinph.2018.08.018

Lisman J., Buzsáki G., Eichenbaum H., Nadel L., Ranganath C., Redish A. D. (2017) **Viewpoints: how the hippocampus contributes to memory, navigation and cognition** *Nature Neuroscience* **20**:1434–1447 https://doi.org/10.1038/nn.4661

Mahmoudzadeh M., Dehaene-Lambertz G., Fournier M., Kongolo G., Goudjil S., Dubois J., Grebe R., Wallois F. (2013) **Syllabic discrimination in premature human infants prior to complete formation of cortical layers** *Proceedings of the National Academy of Sciences of the United States of America* **110**:4846–4851 https://doi.org/10.1073/pnas.1212220110

Mastropieri D., Turkewitz G. (1999) **Prenatal experience and neonatal responsiveness to vocal expressions of emotion** *Developmental Psychobiology* **35**:204–214 https://doi.org/10.1002/(sici)1098-2302(199911)35:3<204::aid-dev5>3.0.co;2-v

May L., Gervain J., Carreiras M., Werker J. F. (2018) **The specificity of the neural response to speech at birth** *Developmental Science* **21** https://doi.org/10.1111/desc.12564

Molnar-Szakacs I., Kupis L., Uddin L. Q. (2021) **Neuroimaging markers of risk and pathways to resilience in autism spectrum disorder** *Biological Psychiatry. Cognitive Neuroscience and Neuroimaging* **6**:200–210 https://doi.org/10.1016/j.bpsc.2020.06.017

Moon C., Lagercrantz H., Kuhl P. K. (2013) **Language experienced in utero affects vowel perception after birth: a two-country study** *Acta Paediatrica* **102**:156–160 https://doi.org/10.1111/apa.12098

Murray S. R., Shenkin S. D., McIntosh K., Lim J., Grove B., Pell J. P., Norman J. E., Stock S. J. (2017) **Long term cognitive outcomes of early term (37-38 weeks) and late preterm (34-36 weeks) births: A systematic review** *Wellcome Open Research* **2** https://doi.org/10.12688/wellcomeopenres.12783.1

Näätänen R., Paavilainen P., Rinne T., Alho K. (2007) **The mismatch negativity (MMN) in basic research of central auditory processing: A review** *Clinical Neurophysiology* **118**:2544–2590 https://doi.org/10.1016/j.clinph.2007.04.026

Nielsen T. M., Pedersen M. V., Milidou I., Glavind J., Henriksen T. B. (2019) **Long-term cognition and behavior in children born at early term gestation: A systematic review** *Acta Obstetricia Et Gynecologica Scandinavica* **98**:1227–1234 https://doi.org/10.1111/aogs.13644

Noble K. G., Fifer W. P., Rauh V. A., Nomura Y., Andrews H. F. (2012) **Academic achievement varies with gestational age among children born at term** *Pediatrics* **130**:e257–264 https://doi.org/10.1542/peds.2011-2157

Partanen E., Kujala T., Näätänen R., Liitola A., Sambeth A., Huotilainen M. (2013) **Learning-induced neural plasticity of speech processing before birth** *Proceedings of the National Academy of Sciences of the United States of America* **110**:15145–50 https://doi.org/10.1073/pnas.1302159110

Paz Levy D., Sheiner E., Wainstock T., Sergienko R., Landau D., Walfisch A. (2017) **Evidence that children born at early term (37-38 6/7 weeks) are at increased risk for diabetes and obesity-related disorders** *American Journal of Obstetrics and Gynecology* **217** https://doi.org/10.1016/j.ajog.2017.07.015

Peña M., Maki A., Kovacić D., Dehaene-Lambertz G., Koizumi H., Bouquet F., Mehler J. (2003) **Sounds and silence: An optical topography study of language recognition at birth** *Proceedings of the National Academy of Sciences of the United States of America* **100**:11702–11705 https://doi.org/10.1073/pnas.1934290100

Picciolini O., Porro M., Meazza A., Gianní M. L., Rivoli C., Lucco G., Barretta F., Bonzini M., Mosca F. (2014) **Early exposure to maternal voice: Effects on preterm infants development** *Early Human Development* **90**:287–292 https://doi.org/10.1016/j.earlhumdev.2014.03.003

Provenzi L., Broso S., Montirosso R. (2018) **Do mothers sound good? A systematic review of the effects of maternal voice exposure on preterm infants' development** *Neuroscience and Biobehavioral Reviews* **88**:42–50 https://doi.org/10.1016/j.neubiorev.2018.03.009

Richard C., Jeanvoine A., Stark A. R., Hague K., Kjeldsen C., Maitre N. L. (2022) **Randomized trial to increase speech sound differentiation in infants born preterm** *The Journal of Pediatrics* **241**:103–108 https://doi.org/10.1016/j.jpeds.2021.10.035

Rose O., Blanco E., Martinez S. M., Sim E. K., Castillo M., Lozoff B., Vaucher Y. E., Gahagan S. (2013) **Developmental scores at 1 year with increasing gestational age, 37–41 weeks** *Pediatrics* **131**:e1475–1481 https://doi.org/10.1542/peds.2012-3215

Rossion B., Jacques C., Jonas J. (2024) **The anterior fusiform gyrus: The ghost in the cortical face machine** *Neuroscience & Biobehavioral Reviews* **158** https://doi.org/10.1016/j.neubiorev.2024.105535

Sato H., Hirabayashi Y., Tsubokura H., Kanai M., Ashida T., Konishi I., Uchida-Ota M., Konishi Y., Maki A. (2012) **Cerebral hemodynamics in newborn infants exposed to speech sounds: A whole-head optical topography study** *Human Brain Mapping* **33**:2092–2103 https://doi.org/10.1002/hbm.21350

Schonhaut L., Armijo I., Pérez M. (2015) **Gestational age and developmental risk in moderately and late preterm and early term infants** *Pediatrics* **135**:e835–841 https://doi.org/10.1542/peds.2014-1957

Sengupta S., Carrion V., Shelton J., Wynn R. J., Ryan R. M., Singhal K., Lakshminrusimha S. (2013) **Adverse neonatal outcomes associated with early-term birth** *JAMA Pediatrics* **167**:1053–1059 https://doi.org/10.1001/jamapediatrics.2013.2581

Singh L., Morgan J. L., Best C. T. (2002) **Infants' listening preferences: Baby talk or happy talk?** *Infancy* **3**:365–394 https://doi.org/10.1207/S15327078IN0303_5

Soderstrom M., Reimchen M., Sauter D., Morgan J. L. (2017) **Do infants discriminate non-linguistic vocal expressions of positive emotions?** *Cognition and Emotion* **31**:298–311 https://doi.org/10.1080/02699931.2015.1108904

Spong C. Y. (2013) **Defining "term" pregnancy: Recommendations from the defining "term" pregnancy workgroup** *JAMA* **309**:2445–2446 https://doi.org/10.1001/jama.2013.6235

Thiede A., Virtala P., Ala-Kurikka I., Partanen E., Huotilainen M., Mikkola K., Leppänen P. H. T., Kujala T. (2019) **An extensive pattern of atypical neural speech-sound discrimination in newborns at risk of dyslexia** *Clinical Neurophysiology* **130**:634–646 https://doi.org/10.1016/j.clinph.2019.01.019

Tita A. T. N. *et al.* (2009) **Timing of elective repeat cesarean delivery at term and neonatal outcomes** *The New England Journal of Medicine* **360**:111–120 https://doi.org/10.1056/NEJMoa0803267

Vaish A., Striano T. (2004) **Is visual reference necessary? Contributions of facial versus vocal cues in 12-month-olds' social referencing behavior** *Developmental Science* **7**:261–269 https://doi.org/10.1111/j.1467-7687.2004.00344.x

Van Lancker D., Cornelius C., Kreiman J. (1989) **Recognition of emotionalLprosodic meanings in speech by autistic, schizophrenic, and normal children** *Developmental Neuropsychology* **5**:207–226 https://doi.org/10.1080/87565648909540433

Vannasing P. *et al.* (2016) **Distinct hemispheric specializations for native and non-native languages in one-day-old newborns identified by fNIRS** *Neuropsychologia* **84**:63–69 https://doi.org/10.1016/j.neuropsychologia.2016.01.038

Virtala P., Putkinen V., Kailaheimo-Lönnqvist L., Thiede A., Partanen E., Kujala T. (2022) **Infancy and early childhood maturation of neural auditory change detection and its associations to familial dyslexia risk** *Clinical Neurophysiology* **137**:159–176 https://doi.org/10.1016/j.clinph.2022.03.005

Walker-Andrews A. S., Grolnick W. (1983) **Discrimination of vocal expressions by young infants** *Infant Behavior and Development* **6**:491–498 https://doi.org/10.1016/S0163-6383(83)90331-4

Wang A. T., Lee S. S., Sigman M., Dapretto M. (2007) **Reading affect in the face and voice: Neural correlates of interpreting communicative intent in children and adolescents with autism spectrum disorders** *Archives of General Psychiatry* **64**:698–708 https://doi.org/10.1001/archpsyc.64.6.698

Webb A. R., Heller H. T., Benson C. B., Lahav A. (2015) **Mother's voice and heartbeat sounds elicit auditory plasticity in the human brain before full gestation** *Proceedings of the National Academy of Sciences of the United States of America* **112**:3152–3157 https://doi.org/10.1073/pnas.1414924112

White-Traut R. C., Nelson M. N., Silvestri J. M., Cunningham N., Patel M. (1997) **Responses of preterm infants to unimodal and multimodal sensory intervention** *Pediatric Nursing* **23**:169–175

Winkler I., Háden G. P., Ladinig O., Sziller I., Honing H. (2009) **Newborn infants detect the beat in music** *Proceedings of the National Academy of Sciences of the United States of America* **106**:2468–2471 https://doi.org/10.1073/pnas.0809035106

Winkler I., Kushnerenko E., Horváth J., Ceponiene R., Fellman V., Huotilainen M., Näätänen R., Sussman E. (2003) **Newborn infants can organize the auditory world** *Proceedings of the National Academy of Sciences of the United States of America* **100**:11812–11815 https://doi.org/10.1073/pnas.2031891100

Wu Y. J., Hou X., Peng C., Yu W., Oppenheim G. M., Thierry G., Zhang D. (2022) **Rapid learning of a phonemic discrimination in the first hours of life** *Nature Human Behaviour* **6** https://doi.org/10.1038/s41562-022-01355-1

Yeung M. K. (2022) **A systematic review and meta-analysis of facial emotion recognition in autism spectrum disorder: The specificity of deficits and the role of task characteristics** *Neuroscience and Biobehavioral Reviews* **133** https://doi.org/10.1016/j.neubiorev.2021.104518

Zhang D., Chen Y., Hou X., Wu Y. J. (2019) **Near-infrared spectroscopy reveals neural perception of vocal emotions in human neonates** *Human Brain Mapping* **40** https://doi.org/10.1002/hbm.24534

Zhang D., Liu Y., Hou X., Sun G., Cheng Y., Luo Y. (2014) **Discrimination of fearful and angry emotional voices in sleeping human neonates: A study of the mismatch brain responses** *Frontiers in Behavioral Neuroscience* **8** https://doi.org/10.3389/fnbeh.2014.00422

Zhao C., Schiessl I., Wan M. W., Chronaki G., Abel K. M. (2021) **Development of the neural processing of vocal emotion during the first year of life** *Child Neuropsychology* **27**:333–350 https://doi.org/10.1080/09297049.2020.1853090

Zimmerman E., Keunen K., Norton M., Lahav A. (2013) **Weight gain velocity in very low-birth-weight infants: Effects of exposure to biological maternal sounds** *American Journal of Perinatology* **30**:863–870 https://doi.org/10.1055/s-0033-1333669

Editors

Reviewing Editor
Daniel Takahashi
Federal University of Rio Grande do Norte, Natal, Brazil

Senior Editor
Barbara Shinn-Cunningham
Carnegie Mellon University, Pittsburgh, United States of America

Reviewer #1 (Public Review):

Summary:
This manuscript aimed to investigate the emergence of emotional sensitivity and its relationship with gestational age. Using an oddball paradigm and event-related potentials, the authors conducted an experiment in 120 healthy neonates with a gestational age range of 35 to 40 weeks. A significant developmental milestone was identified at 37 weeks gestational age, marking a crucial juncture in neonatal emotional responsiveness.

Strengths:
This study has several strengths, by providing profound insights into the early development of social-emotional functioning and unveiling the role of gestational age in shaping neonatal perceptual abilities. The methodology of this study demonstrates rigor and well-controlled experimental design, particularly involving matched control sounds, which enhances the reliability of the research. Their findings not only contribute to the field of neurodevelopment, but also showcase potential clinical applications, especially in the context of autism screening and early intervention for neurodevelopmental disorders.

Comments on the revised version:

After reviewing the authors' response letter and the revised manuscript, I believe they have done a commendable job in addressing my comments.
Additionally, I concur with the concerns raised by Reviewer #2 regarding several potential confounding factors that require better control in their experimental design. These include the differences in physical properties between vocal and nonvocal stimuli, as well as the infant's exposure to the speech/auditory environment. These concerns should be thoroughly and explicitly discussed in the manuscript, ensuring a clearer understanding for the readers.

https://doi.org/10.7554/eLife.95393.2.sa2

Reviewer #2 (Public Review):

This is an important and very interesting report on a change in newborns' neural abilities to distinguish auditory signals as a function of the gestational age (GA) of the infant at birth (from 35 weeks GA to 40 weeks GA). The authors tested neural discrimination of sounds that were labeled 'happy' vs 'neutral' by listeners that represent two categories of sound, either human voices or auditory signals that mimic only certain properties of the human vocal signals. The finding is that a change occurs in neural discrimination of the happy and neutral auditory signals for infants born at or after 37 weeks of gestation, and not prior (at 35 or 36 weeks of gestation), and only for discrimination of the human vocal signals; no change occurs in discrimination of the nonhuman signals over the 35- to 40-week gestational ages tested. The neural evidence of discrimination of the vocal happy-neutral distinction and the absence of the discrimination of the control signals is convincing. The authors interpret this as a 'landmark' in infants' ability to detect changes in emotional vocal signals, and remark on the potential value of the test as a marker of the infants' interest in emotional signals, underscoring the fact that children at risk for autism spectrum disorder may not show the discrimination. Although the finding is novel and interesting, additional discussion is essential so that readers understand two potential caveats affecting this interpretation.

Comments on the revised version:

The revised manuscript does discuss the limitations of the control stimuli, as well as the limitations with regard to conclusions that can be drawn from this data set. I therefore expected the authors to temper a bit their recommendation that this could be a 'screening' signal for autism because these data are not sufficiently strong to make that recommendation. Also, in the same vein, perhaps the title might be adjusted somewhat to suggest less certainty, for example, by using the word "change" rather than "milestone"'? The data are of interest, but the limitations are genuine limitations.

https://doi.org/10.7554/eLife.95393.2.sa1

Author response:

The following is the authors’ response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *More details should be provided in terms of inclusion and exclusion criteria for the participants, as well as missing data due to the non-cooperation of newborns during the experimental process. Potential differences between preterm and full-term infants are worth exploring. Several aspects of EEG data analyses and data interpretation should be better clarified.*
>
> *Here I have several comments and questions to improve the manuscript.*
>
> *(1) It would be wise to know whether there was any missing data due to the non-cooperation of newborns during the experimental process.*

Thank you for the suggestion. While our initial aim was to include 120 neonates in the final data analysis, we actually recruited 198 neonatal participants for this study. The 78 EEG datasets were excluded from the data analysis due to non-cooperation of neonates (n = 75) or technical issues (n = 3). We have incorporated this detailed information in the Subjects subsection (lines 375-383) in the revised manuscript.

> *(2) The authors investigated the impact of gestational age on emotional perceptual sensitivity in newborns by grouping infants of varying gestational ages in the experiment. The methods section mentions that the study conducted experiments within 24 hours after the birth of the newborns. When do preterm infants (with a gestational age of 35 and 36 weeks) begin to exhibit emotional discrimination comparable to full-term newborns?*

This is indeed an intriguing question that merits exploration. However, in our study, we recruited relatively healthy preterm neonates, many of whom were discharged from the hospital with their mothers within 3-5 days after birth. It would have been challenging to arrange for another EEG testing session once these preterm infants reached full-term age, as their parents were unwilling to return to the hospital.

> *(3) When analyzing EEG data, excluding artifacts with peak deviations exceeding ±200 μV is a relatively lenient criterion, potentially resulting in the retention of some large-amplitude artifacts or noise. What is the rationale behind the author's choice of this criterion? Or, in other words, what considerations led to this specific selection?*

In our standard practice, we typically employ a stricter threshold of ±100 μV for artifact removal in studies involving healthy adults and a median threshold of ±150 μV for data from adult patients, such as those with schizophrenia. However, when analyzing neonatal data, we often resort to the loosest criterion of ±200 μV. This decision is primarily due to the inherent challenges associated with neonatal EEG recordings, as we cannot expect newborns to cooperate or remain quiet during the recording process. Consequently, neonatal EEG data tend to contain more artifacts compared to those from healthy adults. Furthermore, the excitability of the newborn brain is notably elevated. This heightened excitability arises from an imbalance in the distribution and function of excitatory and inhibitory neurotransmitter systems. Typically, the expression of excitatory neurotransmitters and their receptors surpasses that of inhibitory neurotransmitters, resulting in increased excitability in the immature brain. This heightened excitability can occasionally lead to the occurrence of paroxysmal electrical activity. As a result, neonatal EEG recordings may at times display large amplitudes, exceeding even 100 μV. In this revision, we have referenced other neonatal/infant EEG studies or technique pipelines that have used the threshold of ±200 μV to support this criterion (lines 483-484).

> *(4) In the Discussion section, the authors mentioned the biomarkers, such as the fusiform gyrus and hippocampus, which have been identified as potential predictors of autism risk. It is suggested that the authors briefly elucidate the crucial role of these biomarkers in processing social information, which would enhance the readability and logicality of this manuscript.*

Thank you for the thoughtful suggestion. We have expanded the discussion concerning the involvement of the fusiform gyrus and hippocampus in social information processing (lines 314-319).

> ***Reviewer #2 (Public Review):***
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> *First, readers need to see spectrograms that show the 0-4000 Hz in more detail, rather than what is now shown (0-10,000 Hz). The vocal signals in clearer spectrograms will show I believe the initial consonant burst and formant frequencies that are unique to human speech and give rise to the perception of the consonant sounds in the vocal signals like 'dada' and 'tutu' that were tested. The control signals will presumably not show these abrupt acoustic changes at their onset, even though they appear (from the oscillograms) to approximate the amplitude envelope. The primary cue distinguishing the happy and neutral signals in both the vocal and control signals is the pitch of the signals (high vs low), but the burst of energy representing the consonants is only contained in the vocal signals; it has no comparable match in the control signals. It is possible that the presence of a sharp acoustic onset (a unique characteristic of consonants in human speech) is especially alerting to the infants, and that this acoustic cue, in the context of the pitch change, enhances discrimination in the vocal case. One*

> *way to test this would be to use only vowel sounds to represent the vocal signals, without consonants.*

Thank you for your expert comments and considerations. We have redrawn Figure 3 using Praat software with a frequency range of 0-5000 Hz, as suggested by Praat's default parameters. Based on the spectrograms, we acknowledge the potential role of consonants in accounting for differences in stimuli. Consequently, we have included this consideration as one of the limitations of our study in this revised version (lines 325-330).

> *Another critical detail that the authors need to include about the signals is an explanation of how the control signals were generated. The text states that the Fo and amplitude envelope of the vocal signals were mimicked in the control signals, but what was the signal used for the controls? Was a pure tone complex modulated, or was pink noise used to generate the control signals? Or were the original vocal signals simply filtered in some way to create the controls, which would preserve the Fo and amplitude envelope? If merely filtered, the control signals still may be perceived as 'vocal' signals, rather than as nonspeech (the Supplement contains the sounds, and some of the control sounds can be perceived, to my ear, as 'vocal' signals).*

We sincerely appreciate your attention to detail regarding the generation of control signals. As a non-specialized laboratory in audio editing, our approach involved filtering the original vocal sounds around the fundamental frequency (f0) and ensuring a balanced mean intensity between vocal and nonvocal stimuli (as now stated in lines 432-437). However, it became evident that certain "vocal" components persisted in the control sounds, particularly noticeable in the sound "tutu". In this revision, we openly acknowledge this oversight (lines 331-333). We extend our gratitude once again for highlighting the importance of meticulous consideration when generating control sounds for a study.

> *Second, there is no information in the manuscript or supplement about the auditory environment of the participants, nor discussion of the fetus' ability to hear in the womb. In the womb, infants are listening to the mothers' bone-conducted speech (which is full of consonant sounds), and we know from published studies that infants can discern differences not only in the prosody of the speech they hear in the womb, but the phonetic characteristics of the mother's speech. The ability at 37 weeks GA or beyond to discriminate the pitch changes in the vocal, but not control signals, could thus be due to additional experience in utero to speech. Another experiential explanation is that the infants born at 37 weeks GA and beyond may be exposed to greater amounts of speech after birth, when compared to those born at 35 and 36 weeks GA, from the attending nurses and from their caregivers, and this speech is also full of consonant sounds. What these infants hear is likely to be 'infant-directed speech,' which is significantly higher in pitch, mirroring the signals tested here. At 37 weeks GA, infants are likely more robust, may sleep less, and are likely more alert. If infants' exposure to speech, either after birth, or their auditory ability to discern differences in speech in utero, is enhanced at 37 weeks GA and beyond, then an 'experience-related' explanation is a viable alternative to a maturational explanation, and should be discussed. Perhaps both are playing a role. As the authors state, many more signals need to be tested to discern how the effect should be interpreted, and other viable interpretations of the current results discussed.*

We acknowledge the importance of considering the auditory environment of participants and the fetus' ability to hear in the womb. In our study, neonates were exposed to a native language environment both before and after birth (as added in lines 385-386), and we took efforts to minimize their exposure to speech stimuli other than those used in the experiment. Specifically, all neonates participated the experiment and underwent EEG recording within the first 24 hours after birth (lines 386-387). They were promptly transported to a dedicated testing room for EEG recording as soon as their condition stabilized after birth. During recording sessions, they were separated from their mothers to minimize exposure to natural speech (as added in lines 459-461). As a result, we believe that both preterm and term neonates were exposed to comparable amounts of speech after birth and before the experiment. We also ensured that all participants were in a natural sleep state during EEG recording. However, it is possible that term neonates slept less and were more attentive to the limited speech stimuli in their environment before the experiment compared to preterm newborns.

The debate surrounding nature versus nurture in neonate and infant development persists. We recognize the potential impact of prenatal auditory experiences on neonatal perceptual sensitivity. Therefore, we have added a brief discussion regarding innate- or experience-related explanations for emotional prosodic discrimination in neonates, aiming to shed light on future research directions (lines 343-351).

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